

Fundamentals of Data Science

Master's Degree in Data Science, Sapienza Università di Roma
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Reference Bibliography

M. P. Deisenroth, A. A. Faisal, C. S. Ong: *Mathematics for Machine Learning*
S. Prince: *Understanding Deep Learning*
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Disclaimer

This document contains notes for the **Fundamentals of Data Science** course, given by **Professors Cinelli and Spinelli** as part of the Master's Degree in Data Science at **Sapienza Università di Roma** during the **first** semester of Academic Year **2026/2027**.

Note: These notes may not cover all details of each lecture, so it is highly suggested to consult the Professor's official resources and materials.

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Contents

1	Introduction to data science	3
1.1	The lifecycle of data science	3
1.2	The role of machine learning in data science	3

Chapter 1

Introduction to data science

1.1 The lifecycle of data science

Data science can be seen as an interdisciplinary field that focuses on extracting actionable insights and knowledge from structured and unstructured data using techniques based on statistical and mathematical methods, as well as computer science algorithms and domain-specific knowledge.

Generally speaking, a data science pipeline starts with understanding the business objectives and translating them into machine learning objectives.

Next, after collecting and cleaning data, the pipeline focuses on selecting and training a model and evaluating the results for deployment, ultimately monitoring model efficacy and restarting the pipeline, either by collecting or data or re-training the model, whenever needed.

1.2 The role of machine learning in data science

Artificial intelligence is a sub-field of computer science that focuses on creating agents that can perform tasks requiring human intelligence and problem solving.

More specifically, Tom Mitchell defines machine learning as “the study of algorithms that improve their performance P at some task T with experience E “, typically focusing on tasks that would be too complex to just hard-code.

Example: Example: Playing checkers

The task $\langle P, T, E \rangle$ to train an agent to play checker can be defined using the following components:

- The agent’s performance P is evaluated as the proportion of matches it wins against an arbitrary opponent.
- The task T to perform is to train the model to make the correct moves during a match.
- The experience E used to train the agent comes from matches the agent plays against itself.

While machine learning and statistics typically use the same instruments and algorithms to discover patterns within data, the two fields are normally considered to be different areas because statistics follows a more rigorous inference-based approach, while machine learning focuses on making predictions using scalable and autonomous techniques.